

Smart-TAP

Oregon State University - AI Capstone 2026

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Background

The Challenge

Complexity of Irrigation Decisions

Farmers face high-stakes decisions relying on various data sources, creating a need for streamlined access to fragmented information to enhance irrigation practices in Oregon



Limitations



- Disconnected datasets create inefficiencies
- Data exists, but it lives in different systems
- Tools are not integrated
- Insights are hard to access

Active Data Sources:

AgriMet - Precipitation, Air Temperature

OpenET - Evapotranspiration

Future Work:

PRISM - Precipitation, Temperature

DRI - Irrigation Efficiency

USDA - Crop Data

OSU Extension Guides - Local
recommendations for farming

Drought.gov - Soil Moisture



AgriMet

Provides local weather data for farmers.



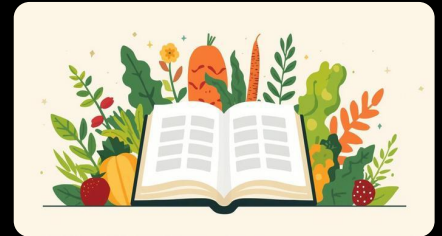
PRISM

Offers climate data to support decision making.



OpenET

Tracks evapotranspiration for water management.



Extension Guides

Offers resources for effective farming practices.

Why This Problem Matters

Irrigation decisions directly impact:

- Water conservation in drought prone regions
- Crop health and yield
- Farm operating costs
- Regional water planning and policy



Existing Tool: Ag Water Chat

<https://agwater.org/agTap>

Existing systems provide:

- **Text-based** answers
- No linked **visualizations**
- Farmers still **switch tabs**, interpret raw datasets, and struggle to create a complete picture.

Prompt:

Submit

rainfall in corvallis last year

I can suggest some sources where you may be able to find precipitation data for Corvallis, Oregon:

1. National Weather Service (NWS) - Corvallis, OR: The NWS website provides historical climate data, including precipitation information for Corvallis.
2. Oregon State University's Climate Services Program: This program provides climate-related information, including precipitation data for Corvallis and other locations in Oregon.
3. National Centers for Environmental Information (NCEI): NCEI is a reliable source of climate data, including precipitation records for Corvallis, OR.

According to the National Weather Service, the average annual precipitation in Corvallis, OR is around 37 inches.

Here are some actual precipitation values for Corvallis, OR for a few recent years:

- 2022: 27.4 inches
- 2021: 31.4 inches
- 2020: 28.5 inches
- 2019: 32.6 inches
- 2018: 24.1 inches

Problem Statement

- Agricultural data is **fragmented** across multiple systems and difficult to access
- Current tools provide **text answers only**, with no linked visualizations
- Farmers and water managers lack a **unified, trustworthy** interface



Our Goal

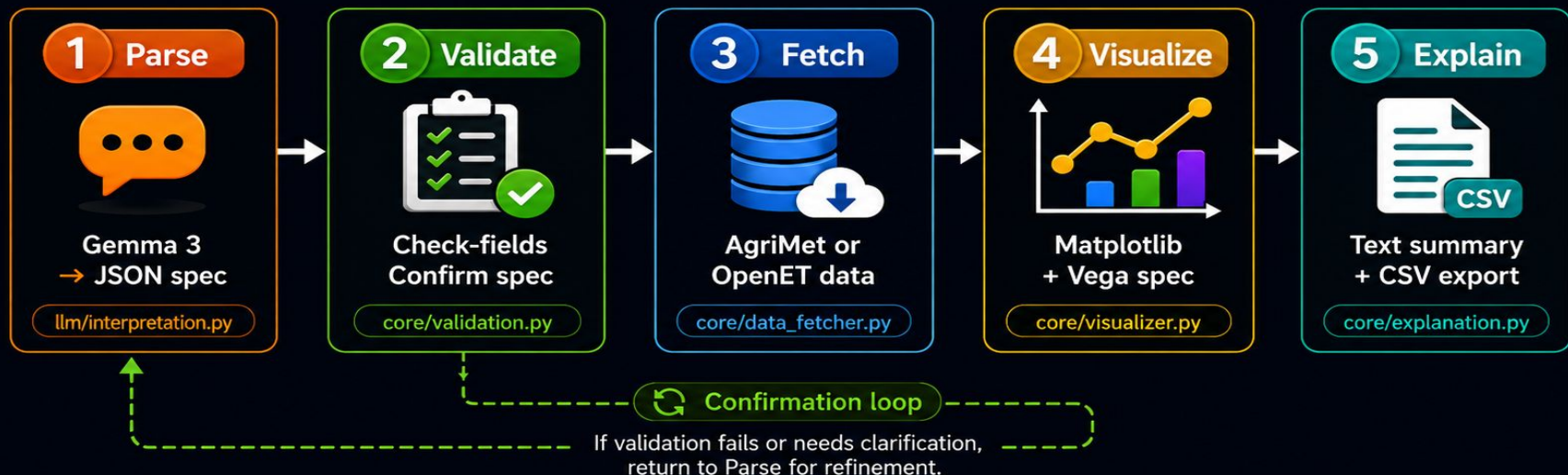
- Our goal is to build a **conversational system** that helps users explore trusted **agricultural** and **climate data**.
- The system **does not** make decisions — it helps users understand the data through visualizations and short explanations.
- Delivers a **single, trusted interface** for Oregon water and climate data



Approach

Approach & Methods

How Smart-TAP Turns a Natural Language Query into a Data Visualization



- **AgriMet:** daily weather station data (**live API**)
- **OpenET:** monthly field-level ET & irrigation (**parquet files**)

Key Design Principle

The LLM only interprets intent, all data retrieval and validation is deterministic and programmatic. No hallucinated data.

Step 1: Plain English Query

Example User Questions

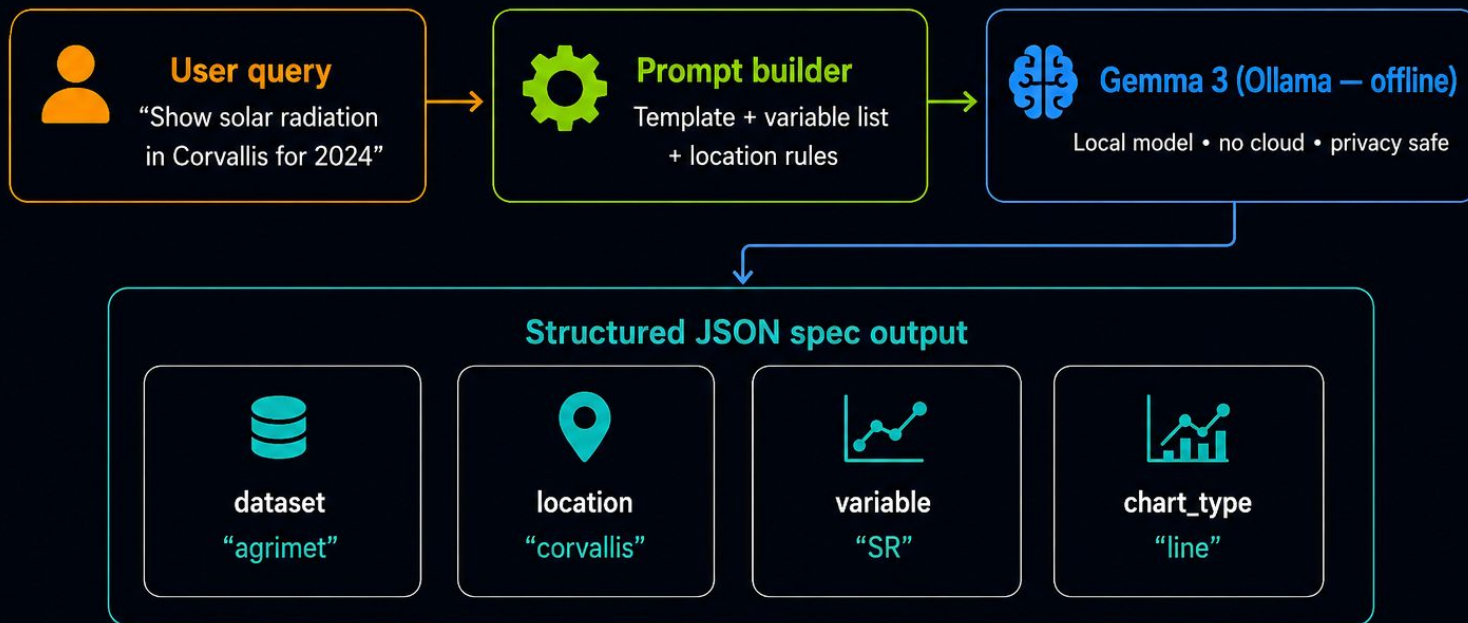
- What crops are grown in Corvallis?
- Show precipitation near Hood River in summer 2024.
- How has crop water use changed for mint fields in Klamath County between 2019 and 2024?
- How many irrigated barley fields were recorded in Baker County from 2016 to 2020?

What SmartTap Accepts

- ▶ Natural-language requests
- ▶ Weather exploration queries
- ▶ Crop and irrigation evidence requests
- ▶ Follow-up edits to existing requests

Users do not need to know dataset names, station IDs, or query syntax.

Step 2 — interpret the query



-  User never needs to know "SR" — they just say "solar radiation".
-  LLM handles the mapping • SmartTap handles validation and defaults.



Prompt snippet:

"Map natural language terms to variable codes. Return only JSON."

Step 2: LLM Interpretation

LLM Interpretation (Gemma)

Model: Local Gemma via Ollama (runs offline)

Input: User's natural language query

Output: Structured JSON spec

Routes to: AgriMet or OpenET dataset

Query: Show solar radiation in Corvallis for 2024

```
{
  "dataset": "agrimet",
  "location": "corvallis",
  "variables": ["SR"],
  "date_range": "2024-01-01 to 2024-12-31",
  "chart_type": "line"
}
```

Data Validation

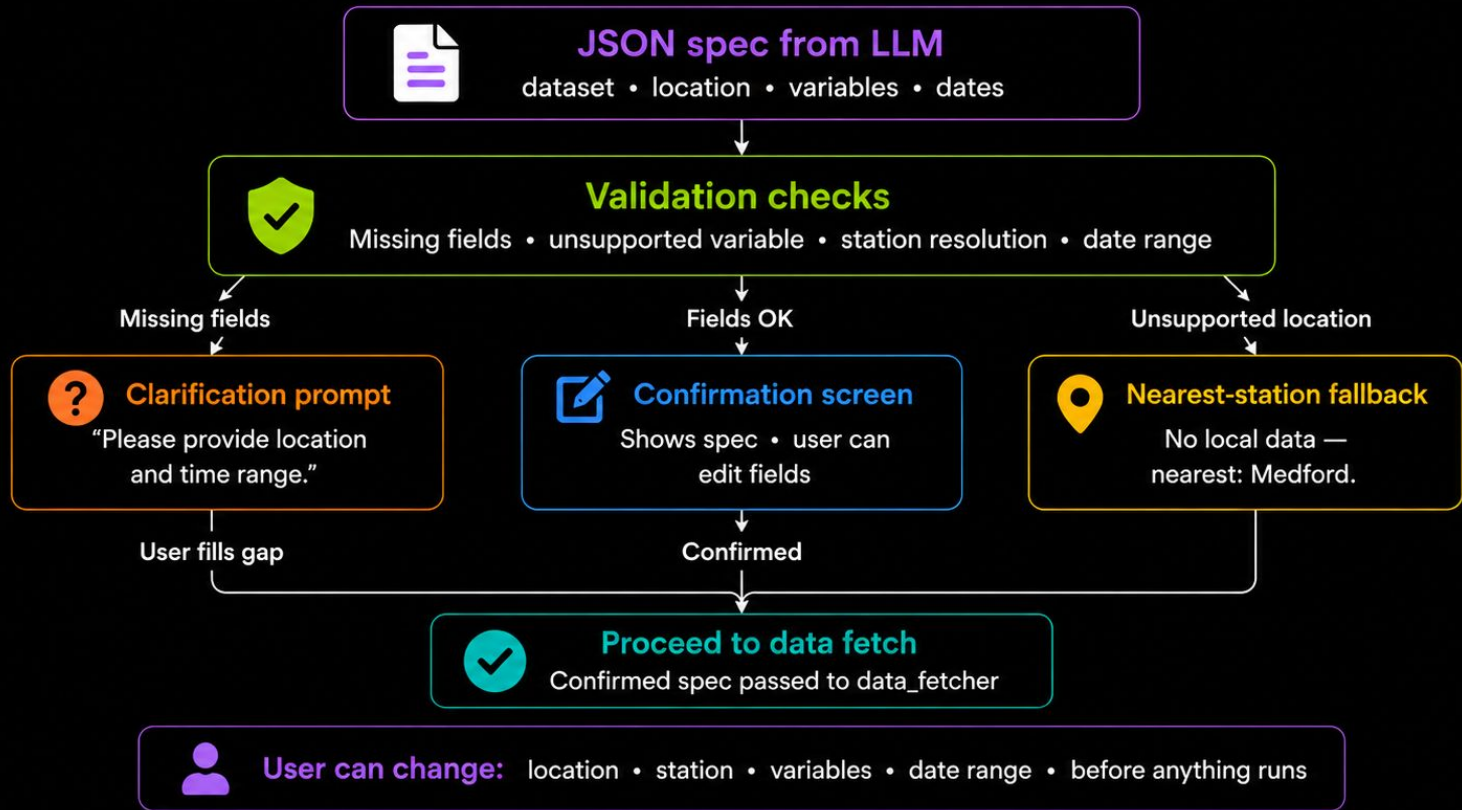
The model handles:

- Intent Classification
- Variable Extraction
- Dataset Selection

SmartTap handles:

- Validation
- Normalization
- Routing
- Defaults
- Execution Logic

Step 3 — Validate and Confirm



Step 3: Validate + Confirm

SmartTap Validates:

- Required fields exist
- Dates are valid
- Variables map to supported metrics
- Dataset/location combinations are usable
- Requests are complete enough to run

Example Validation Checks

- Missing time range
- Unsupported variable
- Unavailable station data
- Invalid crop/location pairing



show temperature



I need a bit more detail before I can build the result. Current request: variables=Average Temperature (deg F), location=the selected location. Please provide the location, the time range.



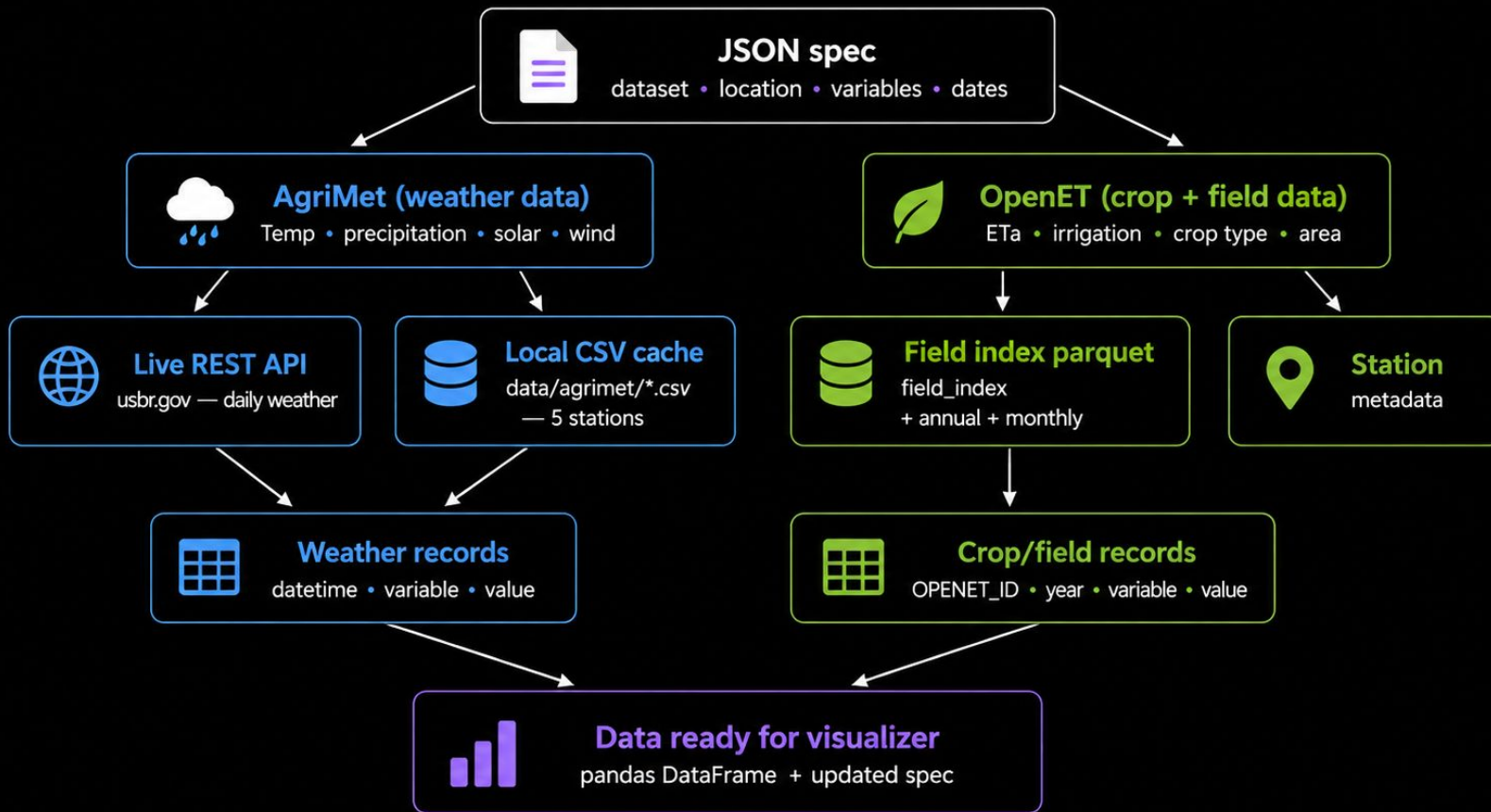
Show temperature in Corvallis for July 2024



Confirm this resolved request before SmartTap runs it:

- Variables: Average Temperature (deg F)
- Location: Corvallis
- Station: Corvallis, Oregon Agrimet Weather Station (crvo)
- Sources: agrimet
- Time range: 2024-07-01 to 2024-07-31

Step 4 — retrieve and check data



Step 4: Retrieve + Check Data

OpenET Retrieval

OpenET Runtime

- Field and crop evidence
- Oregon-focused datasets
- Parquet-backed runtime artifacts
- Crop, location, and time filtering

Supports

- Crop Summaries
- Irrigation Evidence
- Field-level trend exploration

Local GeoPackage → Parquet pipeline

AgriMet Retrieval

AgriMet Runtime

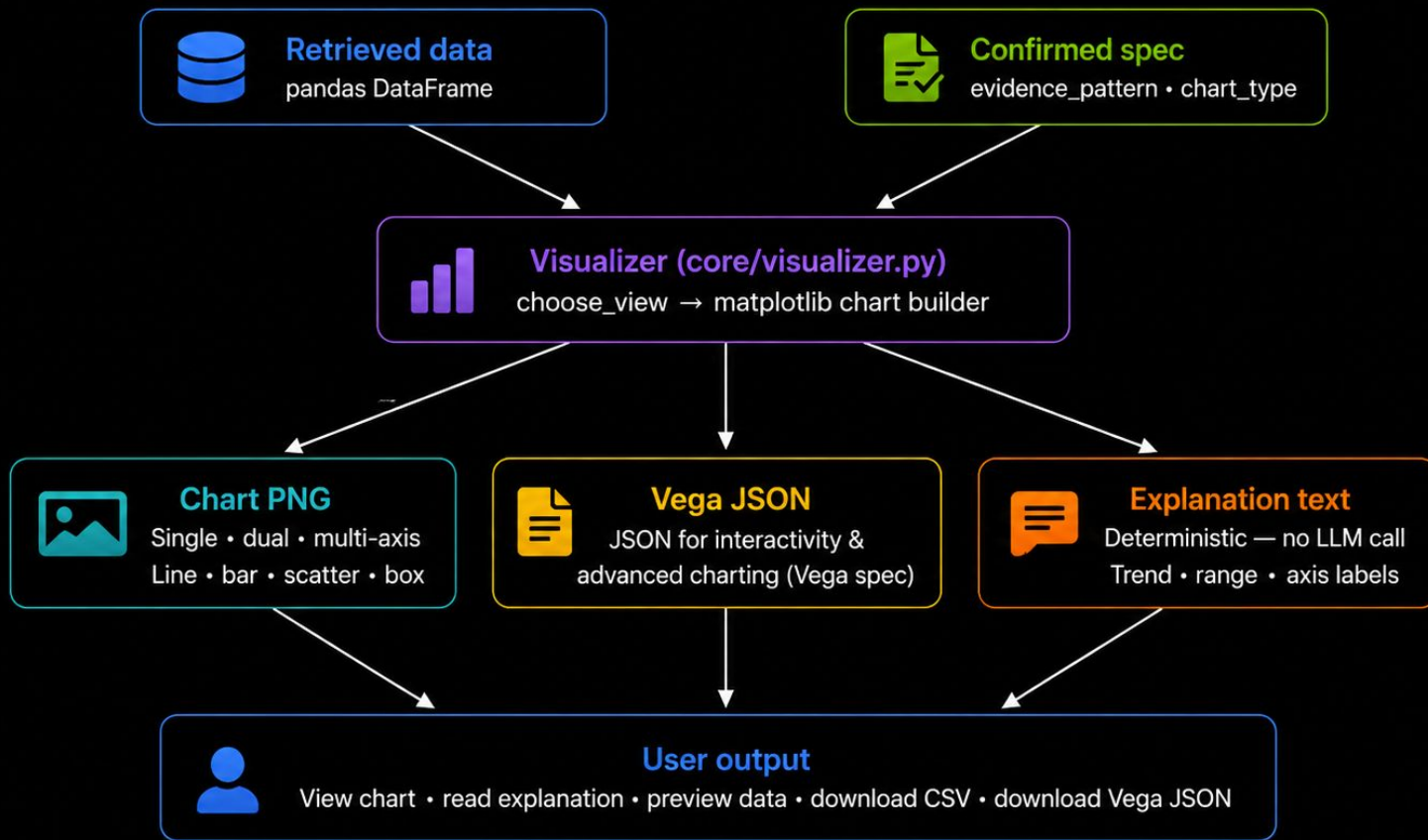
- Weather-station evidence
- Precipitation
- Temperature
- Solar Radiation
- Wind Metrics

Retrieval Features

- Station Resolution
- Variable Availability Checks
- Local/API-backed adapter paths

usbr.gov/pn/agrimet • Live REST API

Step 5 — visualize and explain



Step 5: Evidence Package

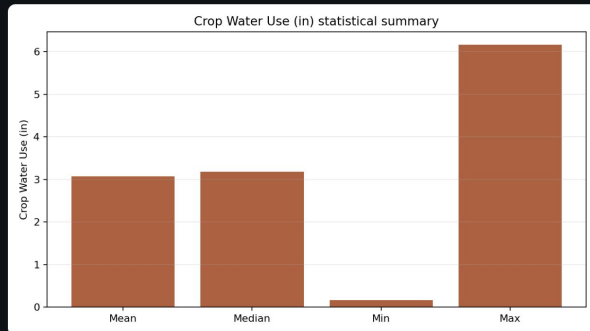
Adaptive Chart Layout

- ▶ Single-axis: similar scales (min/max temp)
- ▶ Dual-axis: mixed scales (rain + solar)
- ▶ Faceted: 3+ variables
- ▶ Vega-Lite spec exported for web embedding

Output Components

- Chart artifacts
- Deterministic summaries
- Validation metadata
- Data previews
- Exportable CSV/evidence outputs

Results



WHAT THIS MEANS

This result summarizes Crop Water Use (in) in Hood River. The chart compares the mean, median, minimum, and maximum across 10 records. The average was 3.073, the median was 3.183, the minimum was 0.166, and the maximum was 6.162.

Download Chart

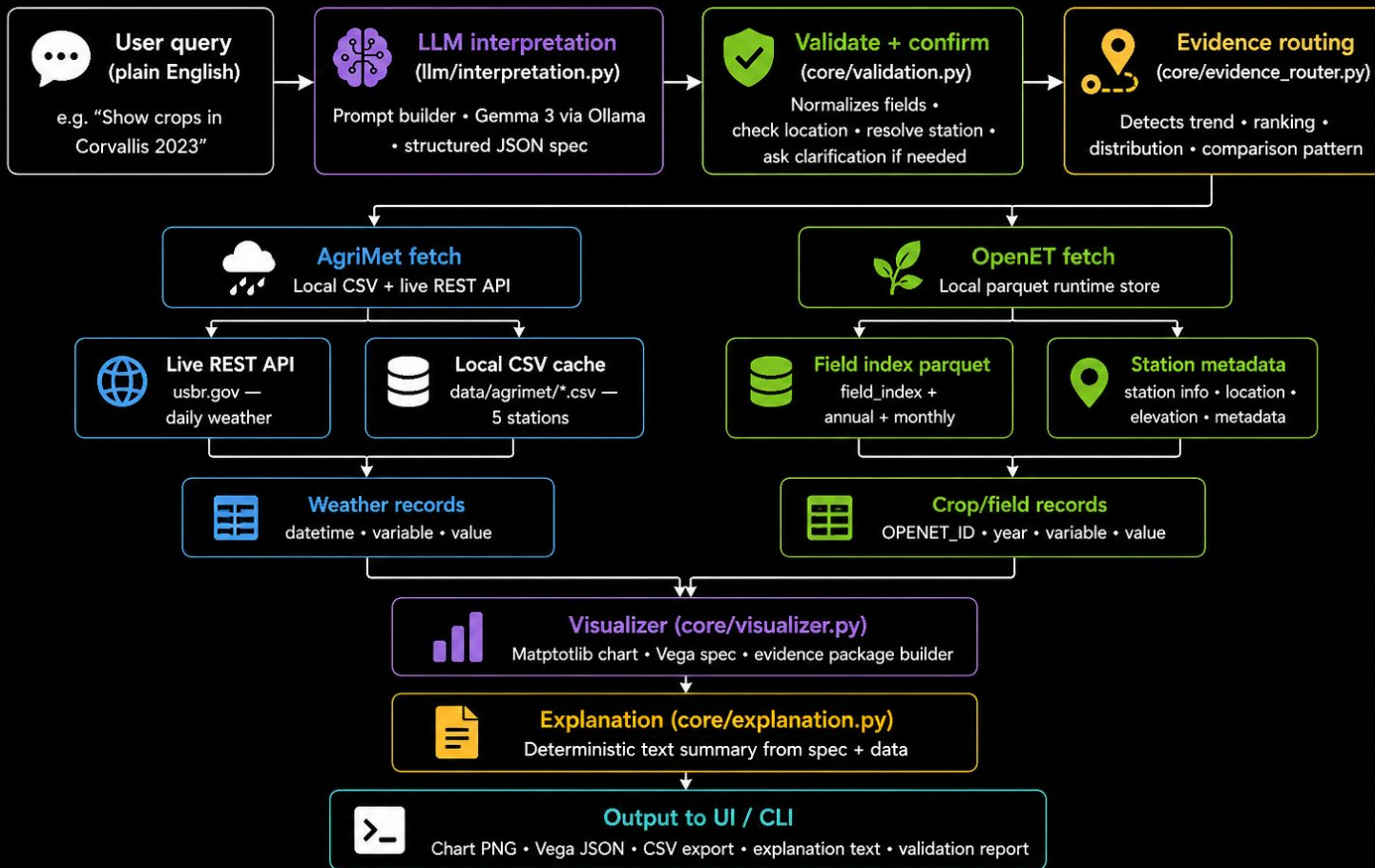
> Result Details

▼ Data Preview

	datetime	ETa
0	2024-01-01 00:00:00	0.166
1	2024-02-01 00:00:00	0.5181
2	2024-03-01 00:00:00	1.8646
3	2024-04-01 00:00:00	3.0806
4	2024-05-01 00:00:00	4.4354
5	2024-06-01 00:00:00	5.3414
6	2024-07-01 00:00:00	6.162
7	2024-08-01 00:00:00	4.5114

Download Data (CSV)

SmartTap — system architecture



Key Features

Intelligent features that go beyond basic data retrieval



Clarification + Confirmation Workflow

Validates incomplete requests, asks clarification questions, and confirms resolved queries before execution.



Multi-Dimensional Charts

Automatically selects single-axis, dual-axis, or faceted layouts based on variables and scale relationships.



Nearby Station Resolver

Resolves nearby AgriMet stations and performs location-aware matching for OpenET exploration.



Confirmation-Based Request Editing

Users can edit resolved requests before execution through a deterministic confirmation workflow.



Crop & Field Exploration

Supports crop distributions, irrigation metrics, and time-series exploration across Oregon field datasets.



Data Validation & Guardrails

Checks payload quality, missing data, invalid ranges, and sparse coverage before generating results.

Results

System Demo - CLI

```
• (venv) (base) mithunkarthikeyan@Mithuns-MacBook-Air SmartTap % python smarttap.py "Show actual evapotranspiration in Corvallis from 2020 to 2022"
```

```
=====
Query: Show actual evapotranspiration in Corvallis from 2020 to 2022
=====
```

```
Parsing query...
```

```
Parsed specification:
```

```
  Dataset: openet
```

```
  Location: Corvallis
```

```
  Variables: ['ETa']
```

```
  Date range: 2020-01-01 to 2022-12-31
```

```
  Chart type: line
```

```
Generating chart...
```

```
Using full Oregon geopackage: preliminary_or_field_geopackage.gpkg
```

```
Querying ETa for Corvallis (city)...
```

```
Querying ETa for 473 fields near Corvallis
```

```
Loaded 36 OpenET records (location)
```

```
Visualization decision: Only one variable requested → single-axis.
```

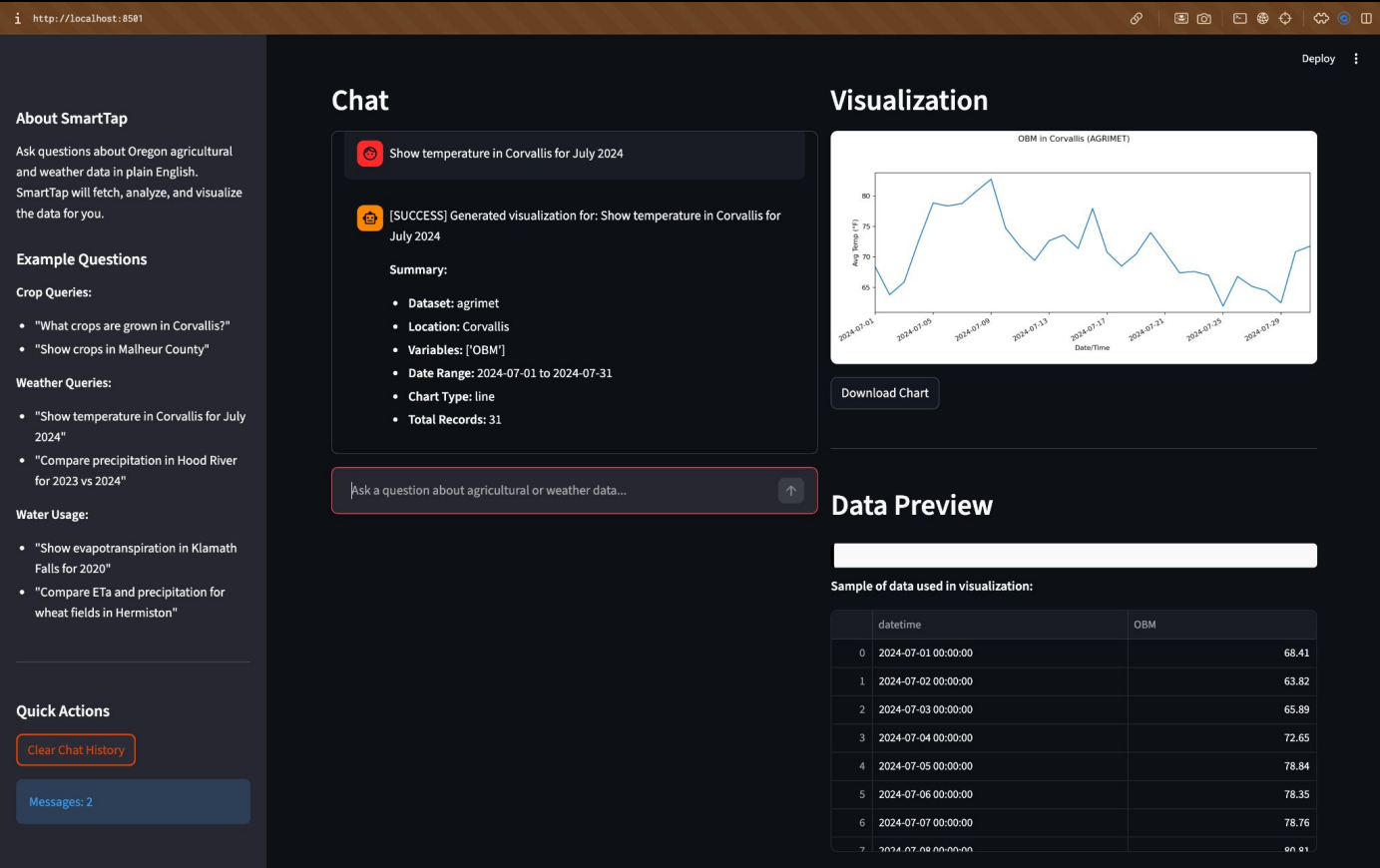
```
PNG: outputs/charts/chart_20260309_010626_chart.png
```

```
Vega: outputs/charts/chart_20260309_010626_chart_vega.json
```

```
  View at: https://vega.github.io/editor/
```

```
=====
Charts generated.
=====
```

System Demo - UI



Evaluation

Evaluation

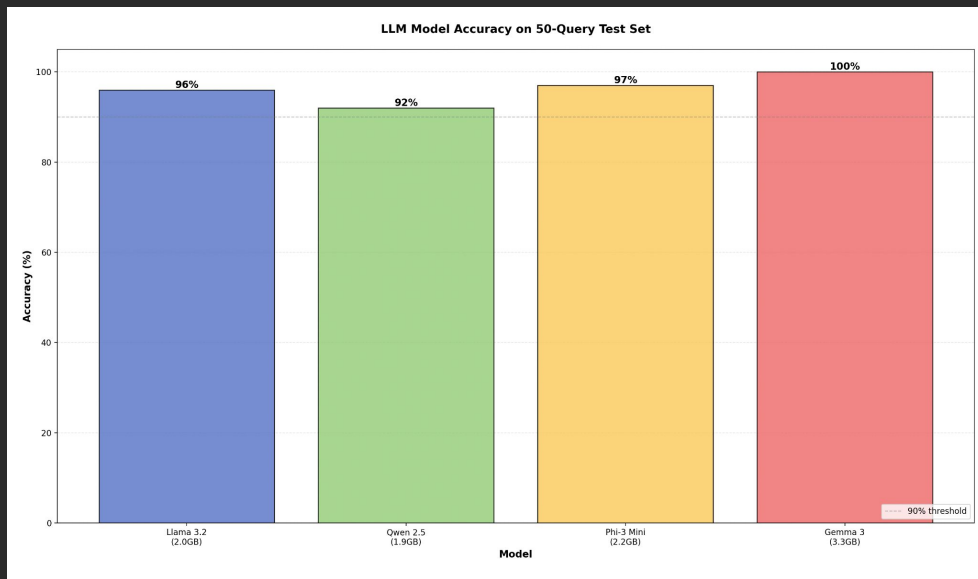
What we Tested:

- **Accuracy:** Does our system correctly answer queries?
- **Speed:** Is our system reasonably fast?
- **Range:** Can our system answer a wide range of queries?

Testing LLM Accuracy across Multiple Models

- Compared 4 local LLM models installed through Ollama
- Ran the same standardized query set across each model
- Measured how accurately each model produced the intended structured request
- Used these results to choose the model that was most reliable for SmartTap parsing

Model	Accuracy
Llama 3.2 (2.0GB)	96%
Qwen 2.5 (1.9GB)	92%
Phi-3 Mini (2.2GB)	97%
Gemma 3 (3.3GB)	100%



Testing Prompt Quality Across Variants

- Compared our full interpretation prompt against a more compact prompt variant
- Used the same retained parser benchmark cases for both prompts
- Measured field-match quality, status accuracy, and stability
- Selected the stronger prompt before using it in the main system

Prompt	Composite Score	Field Match	Status Accuracy	Stability
interpretation.txt	<u>0.9079</u>	0.8955	0.9091	1.0
interpretation_compact.txt	<u>0.8000</u>	0.6667	0.9091	0.9091

Testing Interactive Response Times

- Timed the parsing stage of the pipeline across 4 working local models
- Ran 25+ realistic queries per model
- Compared average parse times to see whether the system felt responsive enough for interactive use
- Used these results to balance model quality against latency

Model	Parse Time (s)
Llama 3.2 (2.0GB)	3.7
Qwen 2.5 (1.9GB)	4.6
Phi-3 Mini (2.2GB)	5.3
Gemma 3 (3.3GB)	4.9

Pipeline Breakdown (using Phi-3):

LLM Parse:	5.3s	
Data Fetch:	1.2s	
Validation:	0.4s	
Visualization:	0.9s	
TOTAL:	7.8s	✓ Under 15s

Testing Query Range with Retained QA Cases

- Ran a retained set of 19 workbook-style queries end-to-end
- Queries selected by project partner as the minimum range to meet
- Covered both OpenET and AgriMet queries across crops, irrigation, weather, and time-series requests
- Ground truth given to us by project advisor Todd to verify against

Result Type	Count
Total retained QA cases	19
Executed-prompt passes	17
Original-prompt passes	15
Rewritten QA prompts	2
Current failures (maybe)	2

Stakeholder Query Evaluation

19 real questions from the project partner workbook — run end-to-end, outputs compared against ground truth

19

Queries run

17

Successful matches

2

Slight differences

17

Exact match (confirmed by independent ground truth)

5

Correct no-data response (data genuinely absent)

2

Correct scope, aggregation under review

Independent ground-truth verification by project advisor Todd McDonnell (in progress). 3 queries confirmed exact; remaining 16 under review.

What Comes Next

- Connect tool with existing AgWater Chat
- Add conversational memory within sessions to support iterative analytical workflows
- Add more datasets related to agriculture and climate into the scope of the project
- Add tool onto AgWater website for user feedback



Thank You

Questions and Discussion Welcome